ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 10	Trace the sequence step by step.
2	**(d)** 14	Pairs: $(1344, 872) \to (872, 472) \to (472, 400) \to (400, 72) \to (328, 72) \to (256, 72) \to (184, 72) \to (112, 72) \to (72, 40) \to (40, 32) \to (32, 8) \to (24, 8) \to (16, 8) \to (8, 8) \to (8, 0)$.
3	**(a)** 6	$2505 = 898 \times 2 + 709$; $898 = 709 \times 1 + 189$; $709 = 189 \times 3 + 142$; $189 = 142 \times 1 + 47$; $142 = 47 \times 3 + 1$; $47 = 1 \times 47 + 0$.
4	**(a)** 2075	The algorithm reverses the digits.
5	**(c)** 12218	LCM $= \dfrac{82 \times 298}{2}$.
6	**(d)** 48	$\lfloor \sqrt{2381} \rfloor = 48$.
7	**(d)** 21	Multiples of 3: 15; of 5: 9; of both: 3.
8	**(d)** 7	$4221 = 3899 \times 1 + 322$; $3899 = 322 \times 12 + 35$; $322 = 35 \times 9 + 7$; $35 = 7 \times 5 + 0$.
9	**(d)** 15	Trace the sequence step by step.
10	**(a)** 14	Pairs: $(1971, 1251) \to (1251, 720) \to (720, 531) \to (531, 189) \to (342, 189) \to (189, 153) \to (153, 36) \to (117, 36) \to (81, 36) \to (45, 36) \to (36, 9) \to (27, 9) \to (18, 9) \to (9, 9) \to (9, 0)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	gcd(320, 294): $320 = 294 \times 1 + 26$; $294 = 26 \times 11 + 8$; $26 = 8 \times 3 + 2$; $8 = 2 \times 4 + 0$ giving 2; gcd(2, 284): $284 = 2 \times 142 + 0$ (1); gcd of the three numbers $= 2$ (1).	1+1
12	Tile side $= \gcd(429, 220)$; $429 = 220 \times 1 + 209$; $220 = 209 \times 1 + 11$; $209 = 11 \times 19 + 0$ (1); Side 11 cm; Number of tiles $= \dfrac{429 \times 220}{11^2} = 780$ (1).	1+1
13	$350 = 346 \times 1 + 4$; $346 = 4 \times 86 + 2$; $4 = 2 \times 2 + 0$; gcd $= 2$ (1); LCM $= \dfrac{346 \times 350}{2} = 60550$ (1).	1+1
14	gcd(844, 854): $854 = 844 \times 1 + 10$; $844 = 10 \times 84 + 4$; $10 = 4 \times 2 + 2$; $4 = 2 \times 2 + 0$ giving 2; gcd(2, 358): $358 = 2 \times 179 + 0$ (1); gcd of the three numbers $= 2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	For $i = 1$ the inner loop runs 11 times, for $i = 2$ it runs 10 times, and so on (1); Count $= 11 + 10 + \cdots + 1 = \frac{11 \times 12}{2}$ (1); Count $= 66$ (1).	1+1+1
16	Squares added: 9, 36, 81 (1); Total $= 9 + 36 + 81$ (1); Final total $= 126$ (1).	1+1+1
17	Sequence: 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 15$ (2).	1+2
	OR: For $i = 1$ the inner loop runs 6 times, for $i = 2$ it runs 5 times, and so on (1); Count $= 6 + 5 + \cdots + 1 = \frac{6 \times 7}{2}$ (1); Count $= 21$ (1).	1+1+1
18	Squares added: 16, 64 (1); Total $= 16 + 64$ (1); Final total $= 80$ (1).	1+1+1
	OR: Sequence: 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 7$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2028 = 1116 \times 1 + 912$; $1116 = 912 \times 1 + 204$; $912 = 204 \times 4 + 96$; $204 = 96 \times 2 + 12$; $96 = 12 \times 8 + 0$ (1); Largest common divisor $= \gcd = 12$ (1); Every common divisor divides 12 (1); Common divisors: 1, 2, 3, 4, 6, 12 (2).	1+1+1+2
	OR: gcd(621, 426): $621 = 426 \times 1 + 195$; $426 = 195 \times 2 + 36$; $195 = 36 \times 5 + 15$; $36 = 15 \times 2 + 6$; $15 = 6 \times 2 + 3$; $6 = 3 \times 2 + 0$ giving 3 (1); gcd(3, 1602): $1602 = 3 \times 534 + 0$ (2); gcd of the three numbers $= 3$ (2).	1+2+2

20	$2175 = 975 \times 2 + 225$; $975 = 225 \times 4 + 75$; $225 = 75 \times 3 + 0$ (1); Largest common divisor $= \gcd = 75$ (1); Every common divisor divides 75 (1); Common divisors: 1, 3, 5, 15, 25, 75 (2).	1+1+1+2
	OR: gcd(2196, 1936): $2196 = 1936 \times 1 + 260$; $1936 = 260 \times 7 + 116$; $260 = 116 \times 2 + 28$; $116 = 28 \times 4 + 4$; $28 = 4 \times 7 + 0$ giving 4 (1); gcd(4, 1236): $1236 = 4 \times 309 + 0$ (2); gcd of the three numbers $= 4$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 5	$2210 = 2161 \times 1 + 49$; $2161 = 49 \times 44 + 5$; $49 = 5 \times 9 + 4$; $5 = 4 \times 1 + 1$; $4 = 1 \times 4 + 0$.
2	**(c)** 6372	The algorithm reverses the digits.
3	**(c)** 960	LCM $= \frac{60 \times 192}{12}$.
4	**(d)** 46	$\lfloor \sqrt{2161} \rfloor = 46$.
5	**(d)** 26	Multiples of 3: 18; of 5: 11; of both: 3.
6	**(d)** 2	$4654 = 4038 \times 1 + 616$; $4038 = 616 \times 6 + 342$; $616 = 342 \times 1 + 274$; $342 = 274 \times 1 + 68$; $274 = 68 \times 4 + 2$; $68 = 2 \times 34 + 0$.
7	**(a)** 9	Trace the sequence step by step.
8	**(b)** 13	Pairs: $(1790, 1060) \to (1060, 730) \to (730, 330) \to (400, 330) \to (330, 70) \to (260, 70) \to (190, 70) \to (120, 70) \to (70, 50) \to (50, 20) \to (30, 20) \to (20, 10) \to (10, 10) \to (10, 0)$.
9	**(c)** 5	$2810 = 2160 \times 1 + 650$; $2160 = 650 \times 3 + 210$; $650 = 210 \times 3 + 20$; $210 = 20 \times 10 + 10$; $20 = 10 \times 2 + 0$.
10	**(d)** 2297	The algorithm reverses the digits.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tile side $= \gcd(480, 265)$; $480 = 265 \times 1 + 215$; $265 = 215 \times 1 + 50$; $215 = 50 \times 4 + 15$; $50 = 15 \times 3 + 5$; $15 = 5 \times 3 + 0$ (1); Side 5 cm; Number of tiles $= \frac{480 \times 265}{5^2} = 5088$ (1).	1+1
12	$510 = 330 \times 1 + 180$; $330 = 180 \times 1 + 150$; $180 = 150 \times 1 + 30$; $150 = 30 \times 5 + 0$; gcd $= 30$ (1); LCM $= \frac{330 \times 510}{30} = 5610$ (1).	1+1
13	gcd(482, 882): $882 = 482 \times 1 + 400$; $482 = 400 \times 1 + 82$; $400 = 82 \times 4 + 72$; $82 = 72 \times 1 + 10$; $72 = 10 \times 7 + 2$; $10 = 2 \times 5 + 0$ giving 2; gcd(2, 612): $612 = 2 \times 306 + 0$ (1); gcd of the three numbers $= 2$ (1).	1+1
14	Tile side $= \gcd(144, 246)$; $246 = 144 \times 1 + 102$; $144 = 102 \times 1 + 42$; $102 = 42 \times 2 + 18$; $42 = 18 \times 2 + 6$; $18 = 6 \times 3 + 0$ (1); Side 6 cm; Number of tiles $= \frac{144 \times 246}{6^2} = 984$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	For $i = 1$ the inner loop runs 5 times, for $i = 2$ it runs 4 times, and so on (1); Count $= 5 + 4 + \cdots + 1 = \frac{5 \times 6}{2}$ (1); Count $= 15$ (1).	1+1+1
16	Squares added: 9, 36, 81 (1); Total $= 9 + 36 + 81$ (1); Final total $= 126$ (1).	1+1+1
17	Sequence: 21, 64, 32, 16, 8, 4, 2, 1 (1); Number of steps $= 7$ (2).	1+2
	OR: For $i = 1$ the inner loop runs 12 times, for $i = 2$ it runs 11 times, and so on (1); Count $= 12 + 11 + \cdots + 1 = \frac{12 \times 13}{2}$ (1); Count $= 78$ (1).	1+1+1
18	Squares added: 9, 36 (1); Total $= 9 + 36$ (1); Final total $= 45$ (1).	1+1+1
	OR: Sequence: 19, 58, 29, 88, 44, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 20$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$1965 = 1485 \times 1 + 480$; $1485 = 480 \times 3 + 45$; $480 = 45 \times 10 + 30$; $45 = 30 \times 1 + 15$; $30 = 15 \times 2 + 0$ (1); Largest common divisor $= \gcd = 15$ (1); Every common divisor divides 15 (1); Common divisors: 1, 3, 5, 15 (2).	1+1+1+2
	OR: gcd(1462, 1830): $1830 = 1462 \times 1 + 368$; $1462 = 368 \times 3 + 358$; $368 = 358 \times 1 + 10$; $358 = 10 \times 35 + 8$; $10 = 8 \times 1 + 2$; $8 = 2 \times 4 + 0$ giving 2 (1); gcd(2, 2188): $2188 = 2 \times 1094 + 0$ (2); gcd of the three numbers $= 2$ (2).	1+2+2

20	$1920 = 760 \times 2 + 400$; $760 = 400 \times 1 + 360$; $400 = 360 \times 1 + 40$; $360 = 40 \times 9 + 0$ (1); Largest common divisor $=$ $\gcd = 40$ (1); Every common divisor divides 40 (1); Common divisors: 1, 2, 4, 5, 8, 10, 20, 40 (2).	1+1+1+2
	OR: gcd(648, 2596): $2596 = 648 \times 4 + 4$; $648 = 4 \times 162 + 0$ giving 4 (1); gcd(4, 476): $476 = 4 \times 119 + 0$ (2); gcd of the three numbers $= 4$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 4452	LCM $= \frac{212 \times 84}{4}$.
2	**(a)** 37	$\lfloor \sqrt{1427} \rfloor = 37$.
3	**(a)** 28	Multiples of 3: 20; of 5: 12; of both: 4.
4	**(d)** 1	$2851 = 2230 \times 1 + 621$; $2230 = 621 \times 3 + 367$; $621 = 367 \times 1 + 254$; $367 = 254 \times 1 + 113$; $254 = 113 \times 2 + 28$; $113 = 28 \times 4 + 1$; $28 = 1 \times 28 + 0$.
5	**(b)** 21	Trace the sequence step by step.
6	**(b)** 14	Pairs: $(1962, 1044) \to (1044, 918) \to (918, 126) \to (792, 126) \to (666, 126) \to (540, 126) \to (414, 126) \to (288, 126) \to (162, 126) \to (126, 36) \to (90, 36) \to (54, 36) \to (36, 18) \to (18, 18) \to (18, 0)$.
7	**(d)** 7	$2951 = 2455 \times 1 + 496$; $2455 = 496 \times 4 + 471$; $496 = 471 \times 1 + 25$; $471 = 25 \times 18 + 21$; $25 = 21 \times 1 + 4$; $21 = 4 \times 5 + 1$; $4 = 1 \times 4 + 0$.
8	**(c)** 9578	The algorithm reverses the digits.
9	**(b)** 3705	LCM $= \frac{65 \times 285}{5}$.
10	**(b)** 41	$\lfloor \sqrt{1697} \rfloor = 41$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$598 = 559 \times 1 + 39$; $559 = 39 \times 14 + 13$; $39 = 13 \times 3 + 0$; gcd $= 13$ (1); LCM $= \frac{559 \times 598}{13} = 25714$ (1).	1+1
12	gcd(452, 522): $522 = 452 \times 1 + 70$; $452 = 70 \times 6 + 32$; $70 = 32 \times 2 + 6$; $32 = 6 \times 5 + 2$; $6 = 2 \times 3 + 0$ giving 2; gcd(2, 158): $158 = 2 \times 79 + 0$ (1); gcd of the three numbers $= 2$ (1).	1+1
13	Tile side $= \gcd(468, 512)$; $512 = 468 \times 1 + 44$; $468 = 44 \times 10 + 28$; $44 = 28 \times 1 + 16$; $28 = 16 \times 1 + 12$; $16 = 12 \times 1 + 4$; $12 = 4 \times 3 + 0$ (1); Side 4 cm; Number of tiles $= \frac{468 \times 512}{4^2} = 14976$ (1).	1+1
14	$627 = 375 \times 1 + 252$; $375 = 252 \times 1 + 123$; $252 = 123 \times 2 + 6$; $123 = 6 \times 20 + 3$; $6 = 3 \times 2 + 0$; gcd $= 3$ (1); LCM $= \frac{375 \times 627}{3} = 78375$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	For $i = 1$ the inner loop runs 8 times, for $i = 2$ it runs 7 times, and so on (1); Count $= 8 + 7 + \cdots + 1 = \frac{8 \times 9}{2}$ (1); Count $= 36$ (1).	1+1+1
16	Squares added: 16 (1); Total $= 16$ (1); Final total $= 16$ (1).	1+1+1
17	Sequence: 30, 15, 46, 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 18$ (2).	1+2
	OR: For $i = 1$ the inner loop runs 9 times, for $i = 2$ it runs 8 times, and so on (1); Count $= 9 + 8 + \cdots + 1 = \frac{9 \times 10}{2}$ (1); Count $= 45$ (1).	1+1+1
18	Squares added: 9, 36, 81 (1); Total $= 9 + 36 + 81$ (1); Final total $= 126$ (1).	1+1+1
	OR: Sequence: 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (1); Number of steps $= 12$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2120 = 689 \times 3 + 53$; $689 = 53 \times 13 + 0$ (1); Largest common divisor $= \gcd = 53$ (1); Every common divisor divides 53 (1); Common divisors: 1, 53 (2).	1+1+1+2
	OR: gcd(2712, 1554): $2712 = 1554 \times 1 + 1158$; $1554 = 1158 \times 1 + 396$; $1158 = 396 \times 2 + 366$; $396 = 366 \times 1 + 30$; $366 = 30 \times 12 + 6$; $30 = 6 \times 5 + 0$ giving 6 (1); gcd(6, 2932): $2932 = 6 \times 488 + 4$; $6 = 4 \times 1 + 2$; $4 = 2 \times 2 + 0$ (2); gcd of the three numbers $= 2$ (2).	1+2+2

20	$1813 = 623 \times 2 + 567$; $623 = 567 \times 1 + 56$; $567 = 56 \times 10 + 7$; $56 = 7 \times 8 + 0$ (1); Largest common divisor $= \gcd = 7$ (1); Every common divisor divides 7 (1); Common divisors: 1, 7 (2).	1+1+1+2
	OR: gcd(1372, 1154): $1372 = 1154 \times 1 + 218$; $1154 = 218 \times 5 + 64$; $218 = 64 \times 3 + 26$; $64 = 26 \times 2 + 12$; $26 = 12 \times 2 + 2$; $12 = 2 \times 6 + 0$ giving 2 (1); gcd(2, 1080): $1080 = 2 \times 540 + 0$ (2); gcd of the three numbers $= 2$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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